Spatial ML - class 5 - Clustering of Geo-located Points

Total points 100/100

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What is the main purpose of clustering geo-located points? * 10/10

- ○ To estimate regression coefficients
- ○ To remove outliers only
- ◉ To divide spatial data into coherent groups
- ○ To calculate distances between all points

In k-means clustering applied to coordinates, what do cluster centers represent? * 10/10

- ○ Random points outside the dataset
- ○ The densest observed points
- ○ Average attribute values only
- ◉ Spatial coordinates representing cluster centers

What is a key difference between k-means and hierarchical clustering? * 10/10

- ◉ They rely on different assumptions and may produce different partitions
- ○ Hierarchical clustering cannot be used for spatial data
- ○ One uses coordinates, the other does not
- ○ One is always more accurate than the other

In hierarchical clustering in R, what does dist() compute? *

10/10

- ( ) Spatial weights matrix
- (•) Matrix of distances between observations
- ( ) Number of clusters
- ( ) Cluster labels

What is the null hypothesis of the Clark-Evans test? *

10/10

- (•) Complete Spatial Randomness (CSR)
- ( ) Points are clustered
- ( ) Points are uniformly distributed
- ( ) Points follow a grid pattern

What does an R index value lower than random indicate in the Clark-Evans test? * 10/10

- ○ Ordered pattern
- ○ No spatial structure
- ◉ Clustered pattern
- ○ Measurement error

In the Clark-Evans test, what is compared? * 10/10

- ◉ Average nearest neighbour distance vs theoretical random distance
- ○ Number of clusters
- ○ Density of raster cells
- ○ Maximum distances between points

What is the main idea behind DBSCAN? *

10/10

- (●) Clustering based on density of points
- ( ) Clustering based on regression residuals
- ( ) Clustering using only centroids
- ( ) Clustering based on hierarchical trees

Which parameters control DBSCAN behaviour? *

10/10

- ( ) k and distance matrix
- (●) eps and minPts
- ( ) number of clusters and iterations
- ( ) bandwidth and kernel

What variables are used in QDC (Quick Density Clustering)? * 10/10

- ○ Cluster labels from DBSCAN
- ○ Regression residuals
- ◉ Distance to neighbours and number of neighbours
- ○ Coordinates only